

GRIDPULSE

Evidence-governed grid intelligence for data-centre development

Technical white paper and development roadmap

Turning fragmented public grid evidence, chronological backtesting and synthetic flexibility studies into a defensible pre-application decision system.

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CORE BOUNDARY

GridPulse does not claim public maps reveal free capacity.

The present platform screens evidence, forecasts published upstream constraint events, and validates capacity and flexibility methods on synthetic networks. Confirmed site-specific MW remains an operator decision.

EXECUTIVE SUMMARY

The investment thesis

Data-centre development increasingly depends on whether large electrical loads can obtain a technically feasible and commercially usable grid connection. Yet early-stage developers face a fragmented evidence problem: public maps describe infrastructure, market platforms publish system-level conditions, and operators hold the decisive topology, loading, queue and protection information. GridPulse is building the evidence and modelling layer between property screening and the formal operator study.

WHAT EXISTS TODAY

A working research platform, not a slide-deck concept.

The separate GridPulse backtest laboratory contains immutable source capture, publication-time event reconstruction, rolling out-of-sample models, AC N-0/N-1 synthetic benchmarks, queue simulations, bounded DeFlex optimization, operator-data contracts and 121 passing tests.

The product wedge

GridPulse can already support public site screening, evidence completeness, upstream constraint history, comparative investigation and customer-defined flexibility scenarios. Its defensible near-term value is reducing uncertainty and improving the quality of connection applications - not replacing the DSO or TSO.

The decisive milestone

The next value inflection is a confidential operator pilot. With anonymized topology, equipment ratings, historical loading, outages, queues and connection outcomes, the same architecture can move from synthetic validation to operator-model validation and historical reconciliation.

Goal	Current evidence state	Readiness
Public site and constraint screening	Public/operator-published context	Available
50Hertz constraint-event risk	Experimental estimate	Research-ready; release gate failed
AC capacity and DeFlex	Synthetic benchmark	Implemented and tested
Operator network analysis	Operator-model unvalidated	Awaiting operator package
Confirmed connection capacity	Operator-confirmed	Unavailable

1. The problem GridPulse addresses

A high-value decision made with incomplete evidence

A data-centre site can be attractive on land, fiber, cooling, permitting and market access while remaining commercially stranded by its grid connection. The relevant decision is not simply whether a transmission line or substation is nearby. It depends on the operator-selected connection point, normal and contingency states, ratings, voltage and protection limits, prior commitments, reinforcement timing and the customer's load behavior.

- Public infrastructure proximity is a screening signal, not proof of electrical connectivity or capacity.
- National load and bidding-zone data describe shared system context, not local substation loading.
- Redispatch and outage publications reveal constraint context, but do not expose residual connection MW.
- A battery or flexible workload can reshape demand; it cannot manufacture network capacity.

Why now

The IEA identifies large, spatially concentrated loads such as data centres as a growing power-system challenge and estimates that grid constraints could delay about 20% of planned global data-centre capacity through 2030. Grid expansion often takes materially longer than data-centre construction, increasing the value of earlier, evidence-based qualification and flexible-connection design.

COMMERCIAL OPPORTUNITY

A faster and more defensible path from site shortlist to operator dialogue.

GridPulse can help developers rank investigation priorities, expose missing evidence, quantify scenario sensitivity, prepare staged or flexible connection requests, and preserve a reproducible record of what was known when each decision was made.

2. Product and evidence architecture

The product is designed as four non-interchangeable evidence layers. Each layer has a distinct source class, permissible language and release gate.

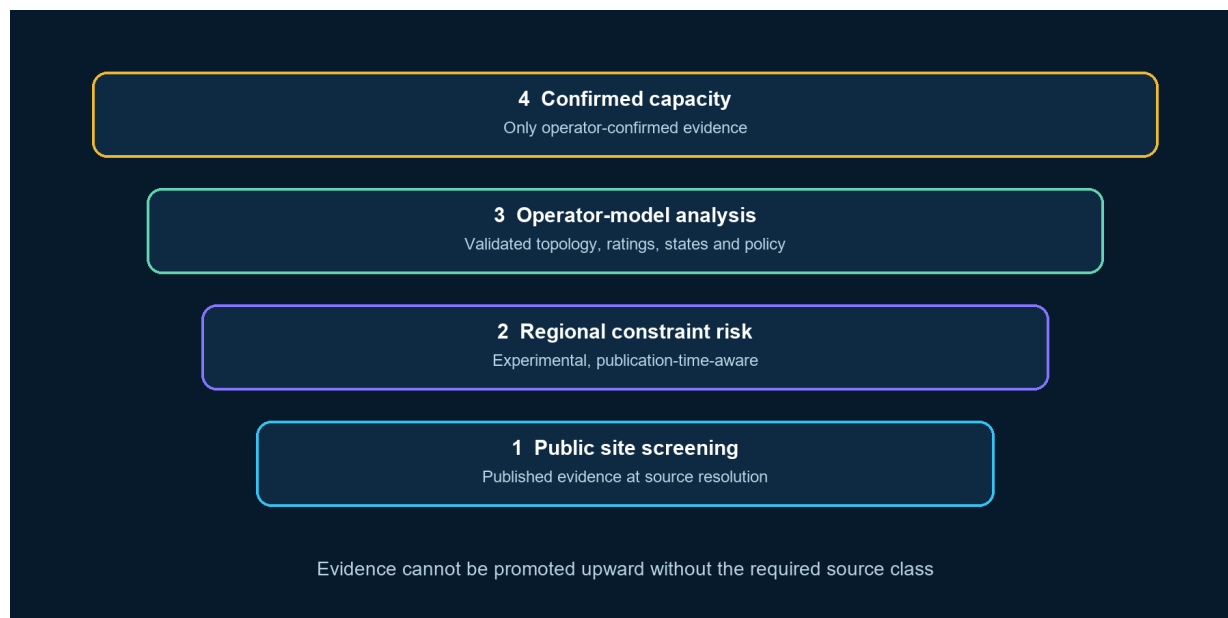


Figure 1. GridPulse evidence stack. Higher layers require stronger evidence; lower-layer results cannot be relabelled as confirmed capacity.

Layer	Permitted output	Permitted interpretation
Public constraint context	Published upstream congestion, outage or planning context	Available MW or connection feasibility
Regional risk	Probability of a published event within a defined horizon	Nodal loading or headroom
Synthetic flexibility	Comparative behavior under explicit mock envelopes	Real Berlin or Brandenburg network result
Operator-confirmed capacity	Operator-reviewed firm or conditional MW	Available until explicit operator evidence exists

3. End-to-end technical system

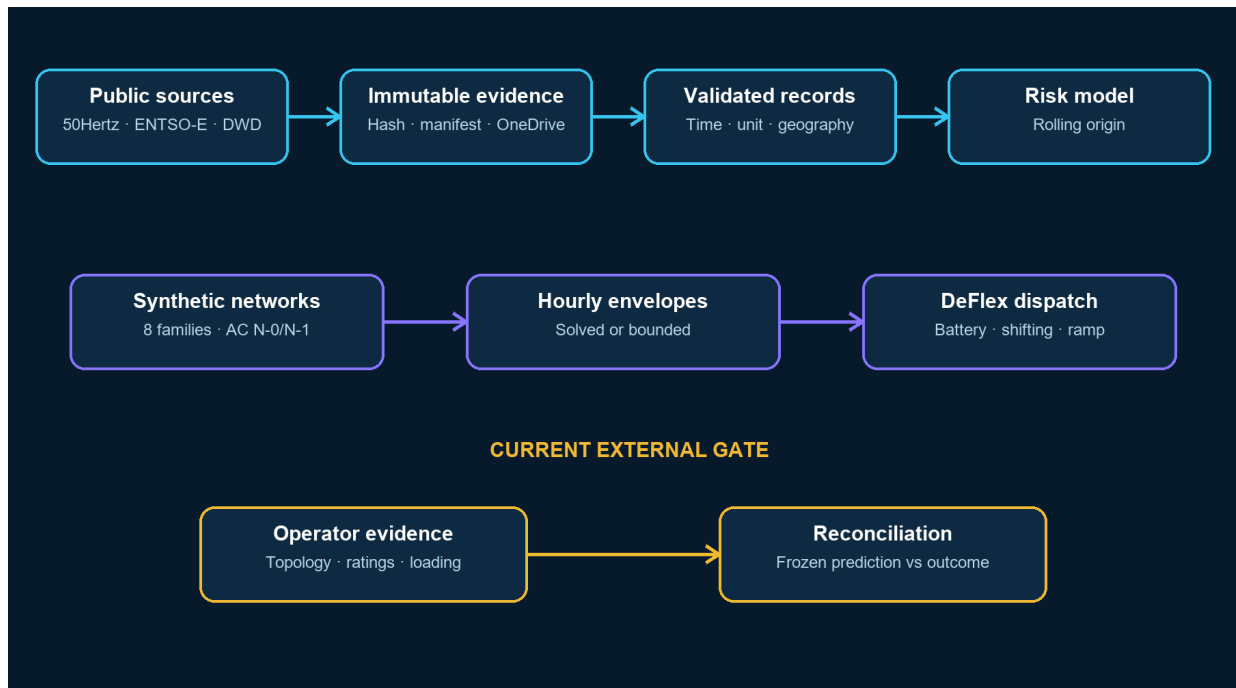


Figure 2. Real public evidence and synthetic electrical studies remain separate until a controlled operator-validation path is available.

3.1 Immutable evidence capture

Downloads are staged locally, hashed, copied atomically to a OneDrive-backed research store and read back for verification. Raw payloads, normalized records, features, models, predictions and simulations occupy separate namespaces. Retrieval observations are preserved even when the underlying payload hash is identical.

Feature	Purpose
SHA-256 content addressing	Detect mutation and avoid duplicate physical payloads
Retrieval manifest	Preserve URL, retrieval time, request parameters and limitations
Raw/normalized separation	Prevent a transformation from overwriting source evidence
OneDrive durable root	Keep large research data out of Git and synchronized to Microsoft cloud storage
Local disposable work root	Keep temporary downloads, caches and active databases out of cloud sync

3.2 Admission before modelling

Availability is not the same as admissibility. A source must establish its publisher, geographic resolution, units, timestamp semantics, revision behavior, missing-versus-zero meaning, licence and allowed uses. If any decisive field cannot be validated, training fails closed.

4. Publication-time-aware regional constraint modelling

4.1 Canonical event contract

The canonical event schema retains source and event identifiers, every observed revision, retrieval and publication times, effective interval, operator, control area, region, affected assets and geography, event type, direction, MW/MWh, reason, status, geographic resolution, identity confidence, truth class, source URL, snapshot hash and validation warnings. Unknown MW and unknown geography remain null.

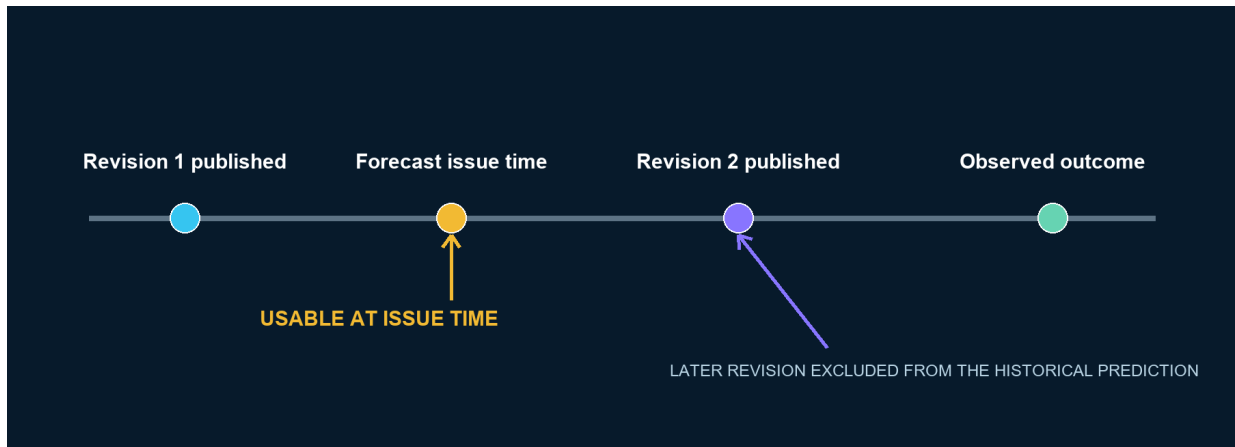


Figure 3. Historical predictions use the latest revision actually published by the issue time; later corrections are excluded.

4.2 Hourly dataset

The current experimental dataset covers the 50Hertz control area from 7 October 2021 through mid-August 2026. Targets represent future non-zero published redispatch intervals, not national-load percentiles. Features are restricted to information available by the forecast issue time.

Forecast horizon	Events	Revisions (events)	Events with missing features
1 hour	42,576	10,307	8,615
6 hours	42,571	15,148	8,614
24 hours	42,553	25,738	8,613
48 hours	42,529	31,849	8,610

4.3 Models and validation

Transparent baselines precede more complex methods: historical event rate, seasonal/calendar rate, logistic regression and a regularized tree baseline. Calibration is fitted only on periods before the held-out period. Whole years are held out in rolling origin; random temporal splits are prohibited.

- Primary rare-event metrics: precision, recall, false-negative rate and PR-AUC.
- Probability quality: Brier score, calibration error and reliability analysis.
- Robustness views: year, season, region/operator, lead time and missing-source periods.

Current interpretation: one- and six-hour yearly precision is comparatively stable in the experimental run, while 24- and 48-hour precision varies materially. Complete publisher-side revision retention, deletion semantics and reuse permission are unresolved, so operational release is not justified.

5. Public data foundation

Source	What contributes here	Resolution	Current limitation
50Hertz congestion management	Quarter-hour increase/reduction and revision chains	TSO control area	Not nodal; archive completeness and reuse unresolved
50Hertz outages	Published event versions and named assets	Transmission asset/context	Rolling feed is not a complete historical archive
50Hertz vertical load	2012-partial 2026 quarter-hour MW	TSO control area	DST folds, missing intervals, publication timing
ENTSO-E Transparency	DE-LU load/generation and latest event documents	Bidding zone / published asset	Superseded document reconstruction not proven
DWD	Historical weather observations and past-only lags	Station / gridded context	Not grid loading or capacity
SMARD / Bundesnetzagentur	National market and redispatch context	Germany / bidding zone	Not local network truth
DSO publications	Planning, procedures and reinforcement context	Service area / named plan	No loading, queues or confirmed headroom

The strongest current real dataset is the 50Hertz congestion-management history: 171,160 unique quarter-hours from 28 September 2021 to 16 August 2026, with 37,612 intervals whose final revision exceeds one. This supports control-area constraint research, not a Berlin-node model.

6. Synthetic AC network benchmark

6.1 Why synthetic networks are necessary

Real operator topology and operating states are unavailable. GridPulse therefore uses explicitly labelled synthetic networks to validate the mechanics of candidate ranking, AC power flow, contingency selection, uncertainty and flexibility. Public map features provide geographic anchors only; electrical impedances, ratings and loading remain mock parameters.

6.2 Electrical method

The benchmark spans eight connected network families and 15 non-slack candidates. For each operating scenario and candidate, pandapower solves AC N-0 and N-1 cases and checks convergence, islanding, bus voltage and equipment thermal limits. Graph algorithms may select likely contingencies, but only the power-flow solver calculates synthetic MW. Exhaustive N-1 cases remain the audit truth.

Network generalization	Leave one network family out	Implemented across eight families
Candidate breadth	All 15 non-slack candidates	Exceeds ten-candidate minimum
Contingency selection	Compare selected cases with exhaustive N-1	V4 recall 100%; 51.21% mean reduction
Hourly approximation	Compare annual proxy with solved hourly AC	95.83% overstatement; gate failed
Truth boundary	All records synthetic_benchmark	No real-network or capacity claim

TECHNICAL LEARNING

A fast proxy is not automatically a safe envelope.

The representative hourly audit found a 41.02 MW mean absolute error and 146.01 MW maximum overstatement for the annual-package approximation. The repository correctly blocks this approximation from becoming a flexibility or connection-capacity boundary.

7. DeFlex: data-centre flexibility under constrained envelopes

7.1 Modelled resources

DeFlex does not apply an arbitrary uplift to an N-1 capacity number. It performs chronological dispatch against an hourly envelope with explicit physical and service constraints.

Input group	Examples
Demand	10, 25, 50, 100 and 200 MW; critical and flexible workload
Interruption	Maximum fraction, consecutive duration and annual hour budget
Recovery	Rebound demand, workload deadline and recovery backlog
Battery	MW, MWh, charge/discharge efficiency, reserve SOC and degradation
Operations	Import ramp, unserved energy and optional backup scenario
Network	Time-varying synthetic N-0/N-1 envelopes and binding constraints

7.2 Optimized dispatch

The latest bounded HiGHS linear-program benchmark evaluated 40 seven-day cases: eight network families by five data-centre sizes. The optimizer produced zero envelope-overstatement hours, shifted 3,375.93 MWh, and explicitly exposed both critical and flexible unserved energy.

	Maximum unserved energy across 168-hour period (MWh)
10 MW	0 MWh
25 MW	0 MWh
50 MW	0 MWh
100 MW	6,112.44 MWh
200 MW	64,522.04 MWh

These values are not Berlin forecasts. Their business meaning is methodological: flexibility can make some constrained envelopes workable, but it cannot close every gap. Large sites remain sensitive to topology, outage state, duration and recovery obligations.

8. Development readiness

GridPulse is beyond concept validation but not yet at operator-grade capacity prediction. The appropriate description is an evidence-governed research prototype with production-relevant data contracts and a completed synthetic workflow benchmark.

Workstream	Level	Evidence	Next goals
Immutable data platform	Advanced prototype	OneDrive storage, hashes, manifests, verification	Long-run monitoring and governance
Public source adapters	Advanced prototype	50Hertz, ENTSO-E, DWD and contextual publications	Licensing and historical completeness
Regional constraint risk	Experimental model	Rolling temporal evaluation at control-area resolution	Admitted archive and stable multi-period release
Synthetic AC benchmark	Validated research method	Eight families, 15 candidates, exhaustive audits	Replace unsafe hourly proxy
Synthetic DeFlex	Validated research method	Chronological bounded dispatch and LP optimization	Real operator envelope and customer profile
Operator-model ingestion	Contract-ready	JSON/CGMES intake and validation gates	First confidential operator package
Operator reconciliation	Workflow-ready	Frozen predictions and outcome schema	Historical operator decisions
Confirmed capacity product	Not available	Correctly blocked	Operator-reviewed connection result

READINESS STATEMENT

What investors should understand.

The platform has de-risked architecture, provenance, chronology, synthetic physics and flexibility mechanics. The remaining uncertainty is not whether code can be written; it is whether decisive operator evidence and validation partnerships can be secured.

9. Data required to reach the next evidence level

9.1 From Stromnetz Berlin, E.DIS and the relevant DSO

- Operator-selected candidate connection point and stable anonymized asset identifiers.
- Anonymized connectivity, normal topology and timestamped switching states.
- Normal and emergency line and transformer ratings, units and validity periods.
- Historical hourly or 15-minute loading and voltage with quality flags.
- Planned and unplanned outages with knowledge time, effective interval and revision.
- Applicable N-1, voltage, thermal, protection and short-circuit policy.
- Contracted demand, accepted-but-not-connected demand, queues and reservations.
- Pseudonymized historical connection requests and their firm, conditional or rejected outcomes.

9.2 From 50Hertz or the relevant TSO

- Upstream network model or approved reduced model, boundary conditions and transfer limits.
- Historic operating states, outage revisions and remedial-action context.
- Security criteria and the interaction between TSO constraints and the DSO connection study.

9.3 From the data-centre developer

- Real hourly or 15-minute IT and facility load profile, ramp and growth plan.
- Critical-load floor, flexible workload, deadlines and service-level penalties.
- UPS/battery telemetry, efficiencies, reserve requirements and degradation economics.
- Backup-generation availability and legal/contractual operating limits.
- Target date, staged MW/MVA request, power factor and minimum acceptable partial capacity.

10. Operator pilot design

The minimum viable pilot does not require public release of confidential network data. It can run inside an operator-approved secure environment with stable random identifiers and agreed geographic aggregation.

Phase	Activity	Output
1. Intake	Validate schema, hashes, units, time zones and permitted uses	Operator-model unvalidated package
2. Historical reconstruction	Rebuild only the topology, outages, loading and queue state known at each issue time	Leakage-safe historical states
3. Frozen predictions	Generate risk and connection-envelope predictions before reading outcomes	Auditable prediction ledger
4. Reconciliation	Compare with actual operator study outcomes and binding constraints	Error, calibration and constraint-match metrics
5. Review	Operator reviews method, caveats and permissible output language	Potential operator-reconciled release

Customer identities, exact topology, loading, queues, protection settings and commercially sensitive connection outcomes would never be published. Public outputs would contain only operator-approved aggregate validation metrics.

11. Product roadmap and investable milestones

M1 Evidence operations	Continuous 50Hertz/ENTSO-E snapshots and consolidated report index	Complete monitoring ledger; no stale-report ambiguity
M2 Public screening product	Regional evidence, risk and missing-data UI	No available-MW claim; reproducible evidence trail
M3 Safe hourly synthetic benchmark	Solved/bounded AC envelopes used by DeFlex	Overstatement and coverage gates pass on unseen families
M4 Operator pilot	Confidential DSO/TSO package and frozen historical predictions	Operator accepts data contract and evaluation protocol
M5 Historical reconciliation	Comparison against connection decisions	Stable results by period and network area
M6 Operator-confirmed workflow	Site-specific reviewed output	Operator explicitly confirms permissible MW statement

Commercialization before operator integration

GridPulse does not need to wait for full DSO integration to create customer value. It can sell evidence-led site screening, comparative investigation, connection-application preparation and customer-defined flexibility analysis. The output must consistently state that available MW is unknown until operator confirmation.

12. Risk register

Risk	Why it matters	Mitigation
Operator access	Decisive data is non-public	Confidential pilot, anonymization and in-boundary execution
False precision	MW outputs can be commercially misleading	Truth classes, null unknowns and release gates
Temporal leakage	Later revisions can inflate historical performance	Publication-time revision reconstruction
Geographic misattribution	Control-area events are not node measurements	Preserve source resolution and identity confidence
Synthetic-to-real gap	Mock physics may not transfer	Leave-family-out tests plus operator reconciliation
Licensing	Public access does not guarantee reuse	Explicit admissibility and permitted-use records
Flexibility overstatement	Power and energy duration are easily confused	Chronological dispatch, SOC, recovery and unserved energy

13. Technical verification record

Check	Audit result
Git state	Clean branch codex/capacity-backtest-foundation at 08d18e4
Unit and fixture tests	121 passed in 40.45 seconds
Ruff lint	Passed
Ruff format	22 pre-existing files would be reformatted
Type checking	No repository type-check command configured
Current synthetic selector	V4 contingency recall 100% in 64 exhaustive audits
Confirmed capacity	Unavailable; available_capacity_mw remains null

14. Conclusion

GridPulse is developing a defensible grid-intelligence layer for large-load development. Its technical differentiation is not a single capacity algorithm. It is the combination of immutable provenance, publication-time reconstruction, fail-closed source admission, temporal backtesting, AC contingency analysis, bounded flexibility optimization and strict separation between public context, experimental risk, synthetic studies and operator-confirmed outcomes.

The platform is ready to support pre-application screening and structured flexibility discussions. It is not yet entitled to state free capacity at a specific location. The fastest credible path to that capability is a confidential pilot with a German DSO and the relevant TSO, using anonymized historical network states and connection outcomes.

INVESTOR TAKEAWAY

The next milestone is evidence access, not a change in scientific principle.

GridPulse has built and tested the research machinery. Capital and partnerships should now concentrate on continuous evidence operations, safe productization, and securing the first operator validation dataset.

Appendix A. Modelling definitions

Term	Definition
N-0	Network operation with all modelled components in service.
N-1	Operation after loss of one relevant component, evaluated through contingency power flow.
Firm envelope	Import level intended to remain feasible under the defined security cases and assumptions.
Conditional envelope	Import permitted only under defined time, topology, curtailment or flexibility conditions.
PR-AUC	Area under the precision-recall curve; useful when constraint events are imbalanced.
Brier score	Mean squared error of probabilistic predictions; lower is better.
Calibration	Agreement between predicted probability and observed frequency.
Truth class	Evidence label controlling how a result may be interpreted or released.
Publication-time leakage	Use of information or revisions that were unavailable when a historical prediction would have been issued.

Appendix B. White-paper design basis

The document structure follows authoritative technical-report principles rather than a commercial downloadable template. NIST technical-series guidance calls for ordered report components and reproducible technical reporting. European Commission evaluation criteria emphasize an executive summary, explicit context and limitations, clear linkage between data, interpretation and conclusions, and technical detail in appendices. This paper adapts those principles to an investor audience while retaining GridPulse's evidence boundaries.

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